DRONE SIMULATION HIJACKING **POSTER**

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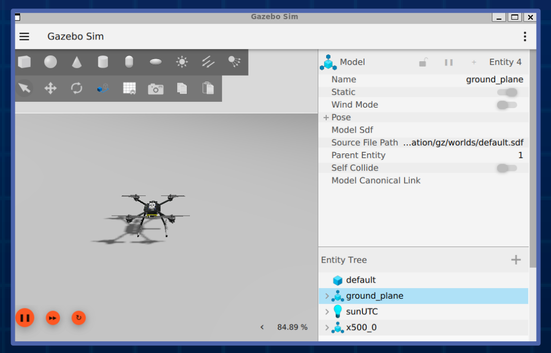
Course Number - 02360125

INTRODUCTION

Our project is about remotely and maliciously attacking a drone and causing it to crash.

The project was done in a simulation environment using PX4, and uses a dictionary attack on the MAVlink protocol. We managed to simulate the hijacking of the key, and the malicious sending of a disarm message to the drone.



METHODOLOGY

01

SETUP

We changed known environment to work with the mavlink protocol.

02

DIVIDE THE PROJECT

PX4 can't properly use Mavlink v2. We divided the attack to two sections: Key recovery, and message injection.

03

EAVESDROPING

Getting a message.

```
[SNIFFER] WARNING: Message COMMAND_LONG is SIGNED! (Key is active)
[SNIFFER] Capturing packet and initiating Dictionary Attack...
[CRACKER] Target signature extracted: bc380af8601f

[SUCCESS] CRITICAL BREACH! Password found: ardupilot
```

04

DICTIONARY ATTACK

Recovering the key from a signed message.

05

INJECTION

Inject a disarm message.

```
shiraz@Shiraz:~/hacker$ python3 inject.py
 Scanning for DRONE (System 1)
 >> LOCKED ON: Drone found! (System 1)
Target Acuired: System 1, Component 1
INJECTING DISARM COMMAND...
 Command Sent. Drone should be falling .
```

CONCLUSION

Using the Mavlink protocol for UAV communication is prone to malicious attacks

ACKNOWLEDGEMENTS

We based our work on "Simulation Framework for GPS Spoofing Attacks" project and "Exploiting the Vulnerabilities in MAVLink Protocol for UAV Hijacking" article.

Project's git – https://github.com/Shiraz-Sh/DroneSimulationHijacking